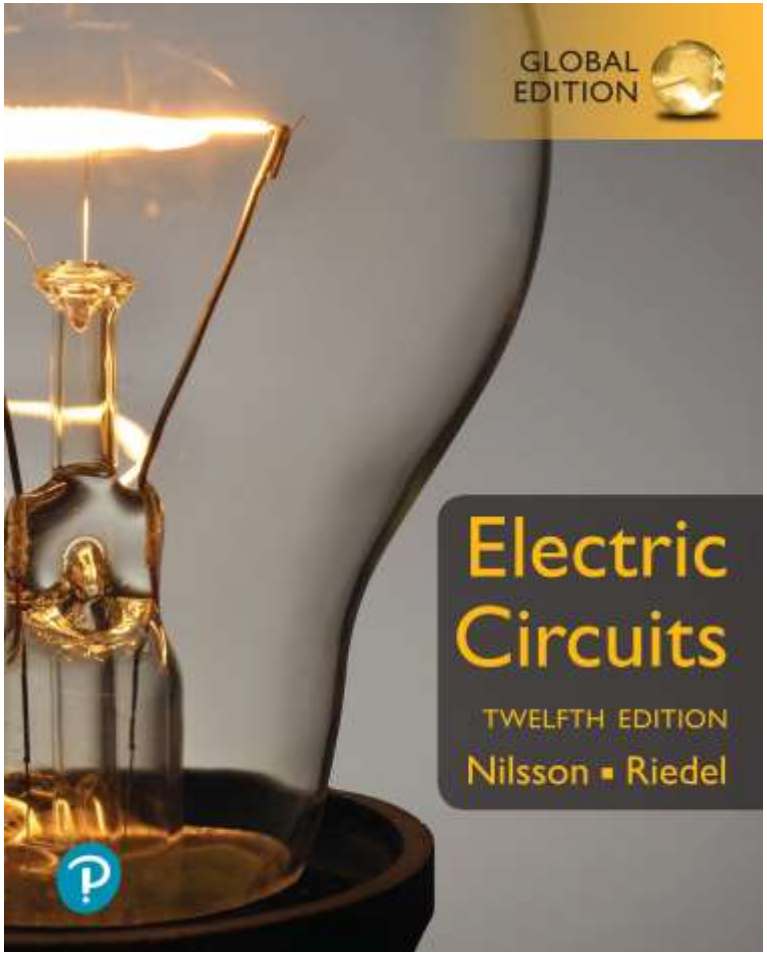
# Electric Circuits

Twelfth Edition, Global Edition




## Chapter 4 – Part 5

Techniques of Circuit Analysis

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

# Student discipline regulations of higher education institutions

Resmi Gazete Tarihi: 18.08.2012 Resmi Gazete Sayısı: 28388

## YÜKSEKÖĞRETİM KURUMLARI ÖĞRENCİ DİSİPLİN YÖNETMELİĞİ

### BİRİNCİ BÖLÜM
### Amaç, Kapsam, Dayanak ve Tanımlar

**Amaç ve kapsam**

**MADDE 1 –** (1) Bu Yönetmeliğin amacı, yükseköğretim kurumları öğrencilerine verilecek disiplin cezaları ile soruşturma usul ve esaslarını düzenlemektir.

(2) Bu Yönetmelik yükseköğretim kurumlarındaki tüm öğrencileri kapsar.

**Dayanak**

**MADDE 2 –** (1) Bu Yönetmelik 4/11/1981 tarihli ve 2547 sayılı Yükseköğretim Kanununun 54 üncü maddesi ile 65 inci maddesinin (a) fıkrasının (9) numaralı bendine dayanılarak hazırlanmıştır.

**Tanımlar**

**MADDE 3 –** (1) Bu Yönetmelikte geçen;

a) Öğrenci: Yükseköğretim kurumlarında önlisans, lisans, yüksek lisans, doktora, tıpta uzmanlık veya sanatta yeterlilik öğrenimi gören kişileri,

b) Kınama: Öğrenciye öğrencilikle ilgili kusurlu davranışlarından dolayı kınandığının yazılı olarak

ARTICLE 7 – (1) Actions that require **suspension** from the higher education institution for **one semester** are as follows;

e) Cheating or having others cheat in exams,

kullandırmak veya başkasına ait bir belgeyi kullanmak,

ç) Yükseköğretim kurumunda kişilerin şeref ve haysiyetini zedeleyen sözlü veya yazılı eylemlerde bulunmak,

d) Yükseköğretim kurumu personelinin, kurum içinde ya da dışında, şeref ve haysiyetini zedeleyen sözlü veya yazılı eylemlerde bulunmak,

e) Yükseköğretim kurumunda alkollü içki içmek,

f) Yükseköğretim kurumuna ait kapalı ve açık mahallerde yetkililerden izin almadan toplantılar düzenlemek.

**Yükseköğretim kurumundan bir yarıyıl için uzaklaştırma cezasını gerektiren disiplin suçları**

**MADDE 7 –** (1) Yükseköğretim kurumundan bir yarıyıl için uzaklaştırma cezasını gerektiren eylemler şunlardır;

a) Yükseköğretim kurumu personeli ve öğrencilerini tehdit etmek,

b) Yükseköğretim kurumlarında işgal ve benzeri fiillerle yükseköğretim kurumunun hizmetlerini engelleyici eylemlerde bulunmak,

c) Kurum personeli ve öğrencilerine fiili saldırıda bulunmak,

ç) Yükseköğretim kurumlarında hırsızlık yapmak,

d) Yükseköğretim kurumu bünyesinde mevcut bina, demirbaş eşya ve benzeri malzemeyi tahrip etmek veya bilişim sistemine zarar vermek,

e) Sınavlarda kopya çekmek veya çektirmek,

f) Seminer, tez ve yayınlarında intihal yapmak.

**Yükseköğretim kurumundan iki yarıyıl için uzaklaştırma cezasını gerektiren disiplin suçları**

**MADDE 8 –** (1) Yükseköğretim kurumundan iki yarıyıl için uzaklaştırma cezasını gerektiren eylemler şunlardır;

a) Yükseköğretim kurumu görevlilerine karşı cebir ve şiddet kullanarak görevin yapılmasına engel olmak.

# Chapter 4 – techniques of circuit analysis
## (1 of 14)

- Node voltage analysis method:
  - Step 1 – Identify each essential node.
  - Step 2 – Pick and label a reference node; then label the node voltages at the remaining essential nodes.
    - If a voltage source is the only element between an essential node and the reference node, replace the node voltage label with the value of the voltage source.
    - **If a voltage source is the only element between two nonreference essential nodes**, combine the nodes and the source into a single supernode.
  - Step 3 – Write a KCL equation for each supernode and for every remaining nonreference essential node where the voltage is unknown.
    - If there are dependent sources, write a constraint equation for each one.
    - If there are supernodes, write a constraint equation for each one.
  - Step 4 – Solve the equations to find the node-voltage values.
  - Step 5 – Solve the circuit using node voltages from Step 4.

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

# Chapter 4 – techniques of circuit analysis
## (2 of 14)

- The mesh current analysis method:
  - Step 1 – Identify the meshes with curved directed arrows that follow the perimeter of each mesh.
  - Step 2 – Label the mesh currents for each mesh.
    - If a current source is in a single mesh, label the mesh current with the value of the current source.
    - If a **current source is shared between two meshes**, **combine the meshes to create a supermesh** and mentally erase the current source.
  - Step 3 – Write a KVL equation for each supermesh and for each single mesh where the current is unknown.
    - If there are dependent sources, write a dependent source constraint equation for each.
    - If there are supermeshes, write a supermesh constraint equation for each.
  - Step 4 – Solve the KVL equations to find the mesh current values.
  - Step 5 – Solve the circuit using mesh currents from Step 4 to find component currents, voltages, and power values.

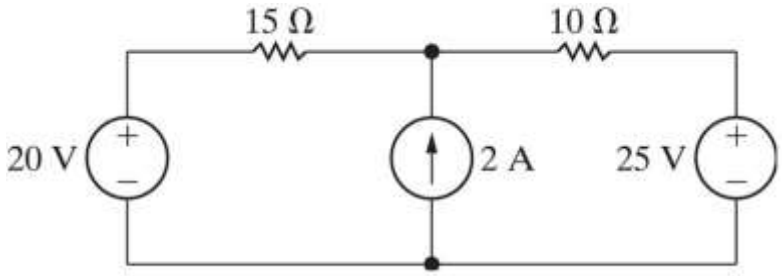
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

# Chapter 4 – techniques of circuit analysis
## (3 of 14)

Node voltage method versus mesh current method

- For the circuit below, find the power for the 2 A source:



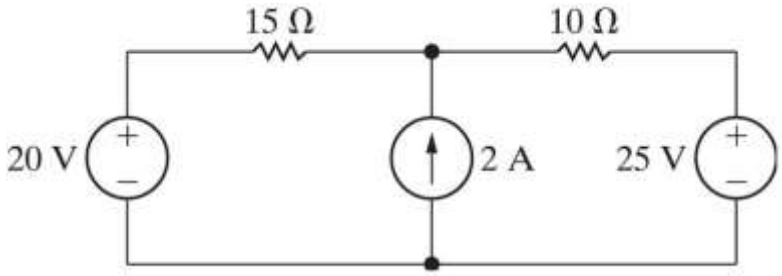

- Node voltage method:
  - Number of essential nodes 2
  - Number of single node KCL equations 1
  - Number of supernode equations 0
  - Number of supernode constraint equations 0
  - Number of dependent source constraint equations 0
  - Equations needed to find desired output 0/1

Pearson
© 2024 Pearson Education Ltd. All Rights Reserved.

# Chapter 4 – techniques of circuit analysis
## (4 of 14)

Node voltage method versus mesh current method

- For the circuit below, find the power for the 2 A source:



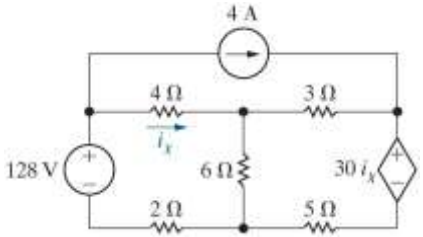

- Mesh current method:
  - Number of meshes 2
  - Number of single mesh KVL equations 0
  - Number of supermesh equations 1
  - Number of supermesh constraint equations 1
  - Number of dependent source constraint equations 0
  - Equations needed to find desired output 1/2

Pearson
© 2024 Pearson Education Ltd. All Rights Reserved.

# Chapter 4 – techniques of circuit analysis
## (5 of 14)

Node voltage method versus mesh current method

- For the circuit below, find the power for the 4 A source:



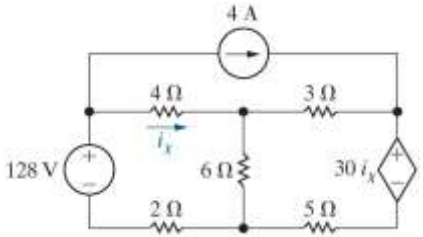

- Node voltage method:
  - Number of essential nodes 4
  - Number of single node KCL equations 3
  - Number of supernode equations 0
  - Number of supernode constraint equations 0
  - Number of dependent source constraint equations 1
  - Equations needed to find desired output 0/1

Pearson
© 2024 Pearson Education Ltd. All Rights Reserved.

# Chapter 4 – techniques of circuit analysis
## (6 of 14)

Node voltage method versus mesh current method

- For the circuit below, find the power for the 4 A source:



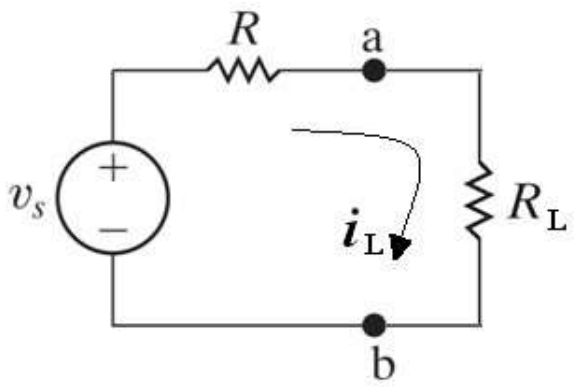

- Mesh current method:
  - Number of meshes 3
  - Number of single mesh KVL equations 2
  - Number of supermesh equations 0
  - Number of supermesh constraint equations 0
  - Number of dependent source constraint equations 1
  - Equations needed to find desired output 1/2

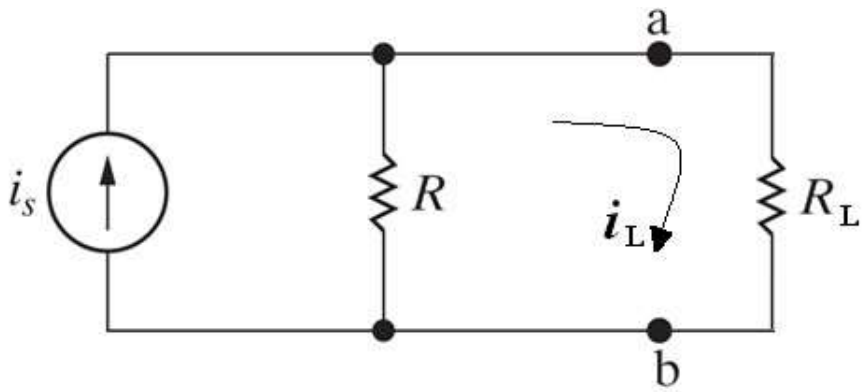
Pearson
© 2024 Pearson Education Ltd. All Rights Reserved.

# Chapter 4 – techniques of circuit analysis
## (7 of 14)

Source transformation

- Under what condition(s) are the two circuits below “equivalent”? “Equivalent” means that if you attached any resistor to both circuits, it would have the same current, and therefore the same voltage and power.




$$i_L = \frac{v_S}{R + R_L}$$

$$i_L = \frac{R \,||\, R_L}{R_L} i_S = \frac{RR_L / R + R_L}{R_L} i_S = \frac{Ri_S}{R + R_L}$$

$\Rightarrow \quad v_S = Ri_S$ - the condition for source transformation

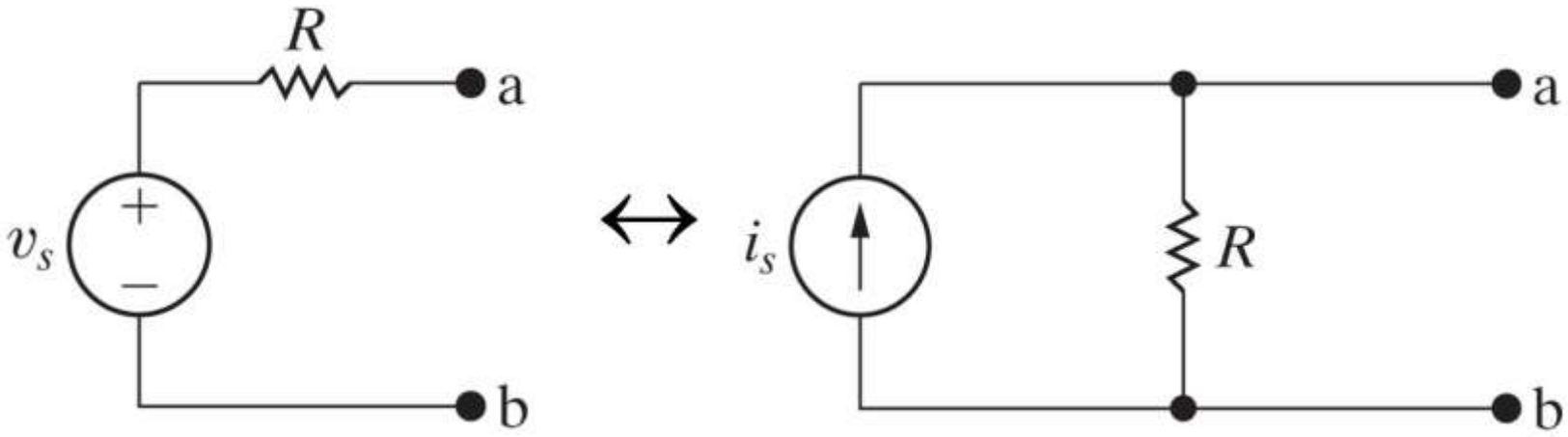
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

# Chapter 4 – techniques of circuit analysis
## (8 of 14)

The condition for source transformation is $v_s = Ri_s$.

simply Ohm's Law.



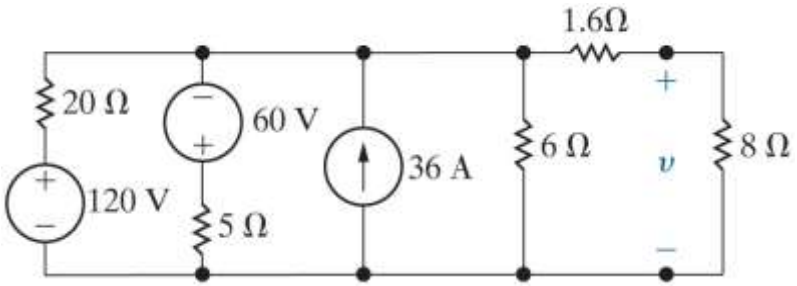

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

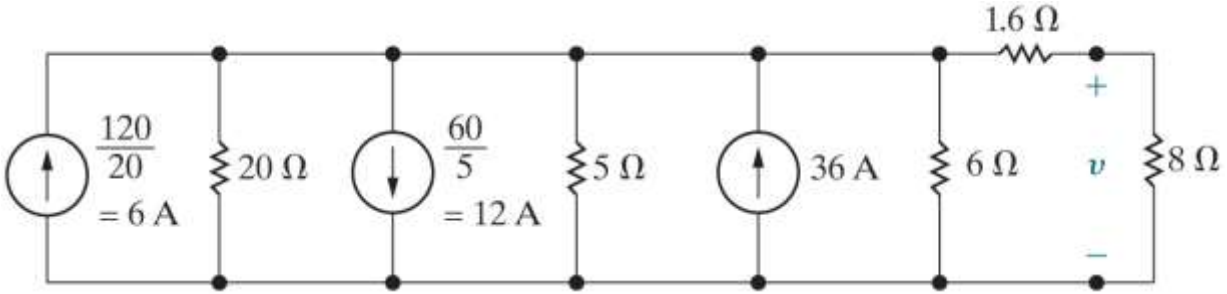
# Chapter 4 – techniques of circuit analysis
## (9 of 14)

Source transformation

- Find *v* in the circuit below by using a sequence of source transformations.




First step – source transform the two left-most branches in the circuit from series voltage source and resistor to parallel current source and resistor:

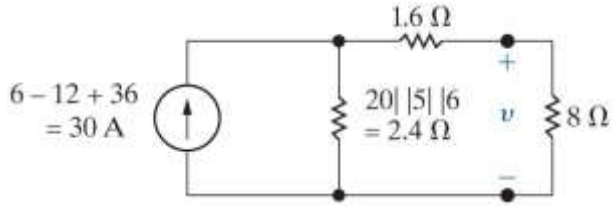



Pearson

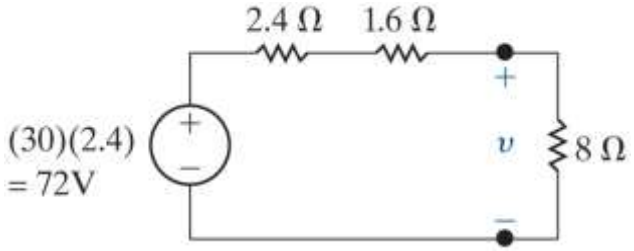
© 2024 Pearson Education Ltd. All Rights Reserved.

# Chapter 4 – techniques of circuit analysis
## (10 of 14)

Second step – combine the three parallel-connected current sources into one current source, using KCL, and the three parallel-connected resistors into one resistor:




Third step – source transform the parallel current source and resistor to a series voltage source and resistor:




Fourth step – use voltage division to find $v$:

$$v = \frac{8}{2.4 + 1.6 + 8}(72) = 48 \text{ V}$$

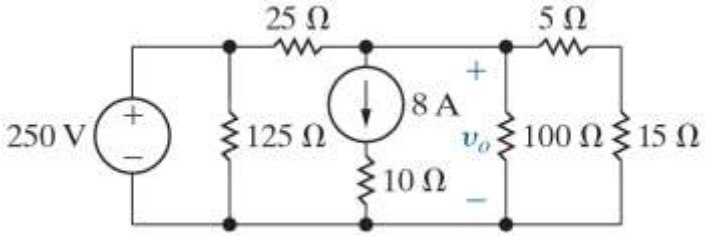
Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

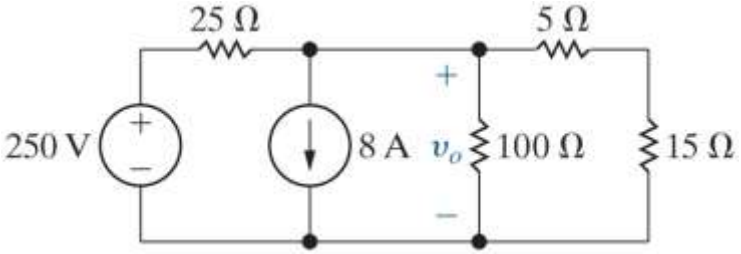
# Chapter 4 – techniques of circuit analysis
## (12 of 14)

Source transformation

- Find $v_o$ and the source power in the circuit below by using a sequence of source transformations.




First step – remove the resistor in parallel with the 250 V source, since it does not change the voltage drop between the top and bottom nodes on the left. Remove the resistor in series with the 8 A current source, since it does not change the current in that branch:




Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

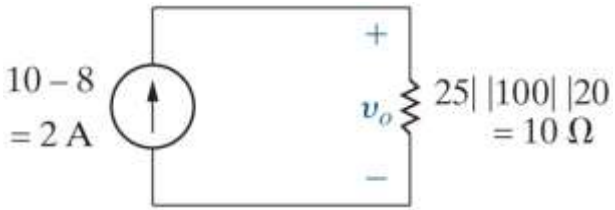
# Chapter 4 – techniques of circuit analysis
## (13 of 14)

Second step – source transform the series-connected voltage source and resistor on the left, and combine the two series-connected resistors on the right into one equivalent resistor:

Third step – combine the two parallel current sources into one source, using KCL, and the three parallel resistors into one resistor:

Fourth step – use Ohm's Law to find $v_o$:

$$v_o = (2)(10) = 20 \text{ V}$$

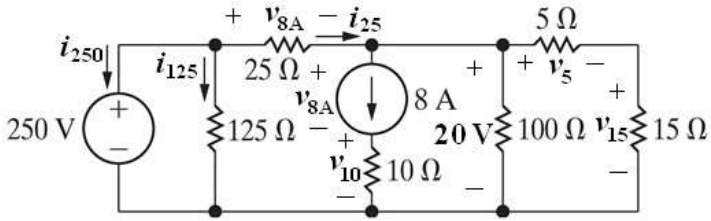



Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

# Chapter 4 – techniques of circuit analysis
## (14 of 14)

Fifth step – to balance the power, reinsert the resistors removed in Step 1:




$$v_{15} = \frac{15}{20}(20) = 15 \text{ V}; \quad v_5 = 20 - 15 = 5 \text{ V}$$

$$v_{10} = (8)(10) = 80 \text{ V}; \quad v_{8A} = 20 - 80 = -60 \text{ V}$$

$$v_{25} = 250 - 20 = 230 \text{ V}; \quad i_{25} = 230/25 = 9.2 \text{ A}$$

$$i_{125} = 250/125 = 2 \text{ A}; \quad i_{250} = -2 - 9.2 = -11.2 \text{ A}$$

$$p_{250} = (250)(-11.2) = -2800 \text{ W}; \quad p_{125} = 250^2/125 = 500 \text{ W}$$

$$p_{25} = (9.2)^2(25) = 2116 \text{ W}; \quad p_{8A} = (8)(-60) = -480 \text{ W}$$

$$p_{10} = (8)^2(10) = 640 \text{ W}; \quad p_{100} = 20^2/100 = 4 \text{ W}$$

$$p_5 = (5)^2/(5) = 5 \text{ W}; \quad p_{15} = 15^2/15 = 15 \text{ W}$$

$$-2800 + 500 + 2116 - 480 + 640 + 4 + 5 + 15 = 0$$

Pearson

© 2024 Pearson Education Ltd. All Rights Reserved.

# Copyright



**This work is protected by United Kingdom copyright laws and is provided solely for the use of instructors in teaching their courses and assessing student learning. Dissemination or sale of any part of this work (including on the World Wide Web) will destroy the integrity of the work and is not permitted. The work and materials from it should never be made available to students except by instructors using the accompanying text in their classes. All recipients of this work are expected to abide by these restrictions and to honor the intended pedagogical purposes and the needs of other instructors who rely on these materials.**



© 2024 Pearson Education Ltd. All Rights Reserved.